Conceptual question · Unit 5 · 2 points · 15 min · Conceptual / fast

C05 — Work of inertia forces: the Coriolis force on a rotating rod and a free particle in an accelerating frame

> *S_0 is an inertial frame. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.*

Part (a) · 1 point

Bead on a rotating rod

A straight, smooth rod lies in a horizontal plane and rotates about a fixed vertical axis through its end O with constant angular velocity $\boldsymbol{\Omega} = \Omega\,\mathbf{k}$ ($\mathbf{k}$ upward). A bead of mass m slides on the rod; r is its distance to O, and $S_1 = \{O; \mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{k}\}$ is the frame of the rod ($\mathbf{e}_r$ along the rod). The bead is released at rest relative to the rod at $r = r_0$. Statement: "In S_1 the Coriolis force does work on the bead, and this work is what makes the relative kinetic energy of the bead grow as it slides outwards."

Step 1. Forces in S_1

S_1 rotates with constant $\boldsymbol{\Omega}$ about the fixed point O, so (Notes §4.9) $m\mathbf{a}_{rel} = \mathbf{F} + \mathbf{F}_{cf} + \mathbf{F}_{cor}$ with the centrifugal force $\mathbf{F}_{cf} = -m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times r\,\mathbf{e}_r) = m\Omega^2 r\,\mathbf{e}_r$ and the Coriolis force

$$\mathbf{F}_{cor} = -2m\,\boldsymbol{\Omega} \times \mathbf{v}_{rel} = -2m\,\Omega\,\mathbf{k} \times \dot{r}\,\mathbf{e}_r = -2m\Omega\dot{r}\,\mathbf{e}_\varphi$$

(no Euler term since $\dot{\boldsymbol{\Omega}} = \mathbf{0}$). The real forces are the weight and the rod reaction, both orthogonal to $\mathbf{e}_r$ on a smooth rod.

Step 2. Powers

With $\mathbf{v}_{rel} = \dot{r}\,\mathbf{e}_r$: $\mathbf{F}_{cor} \cdot \mathbf{v}_{rel} = -2m\Omega\dot{r}^2\,(\mathbf{e}_\varphi \cdot \mathbf{e}_r) = 0$. This holds for any motion, because $\boldsymbol{\Omega} \times \mathbf{v}_{rel}$ is perpendicular to $\mathbf{v}_{rel}$: **the Coriolis force never works in the moving frame.** Only the centrifugal force works, with power $m\Omega^2 r\dot{r}$.

Step 3. Relative kinetic energy

Along $\mathbf{e}_r$: $m\ddot{r} = m\Omega^2 r$, whose solution with $r(0) = r_0$, $\dot{r}(0) = 0$ is $r = r_0\cosh\Omega t$. Work-energy theorem in S_1, with the work of the centrifugal force from r_0 to r given by Barrow's rule:

$$\tfrac{1}{2}m\dot{r}^2 - 0 = \int_{r_0}^{r} m\Omega^2 s\,ds = \left[\tfrac{1}{2}m\Omega^2 s^2\right]_{r_0}^{r} = \tfrac{1}{2}m\Omega^2\left(r^2 - r_0^2\right)$$

The growth comes from the centrifugal force; the Coriolis force only presses the bead against the rod, where the reaction $N\mathbf{e}_\varphi + N_z\mathbf{k}$ balances it: along $\mathbf{e}_\varphi$, $N = 2m\Omega\dot{r}$ against the Coriolis force, and along $\mathbf{k}$, $N_z = mg$ against the weight $-mg\,\mathbf{k}$ (both components are orthogonal to the smooth rod). The statement is FALSE.

Step 4. Seen from S_0

In S_0 there are no inertia forces; the energy is supplied by the rod reaction $N\mathbf{e}_\varphi + mg\,\mathbf{k}$, whose power with the absolute velocity $\mathbf{v} = \dot{r}\,\mathbf{e}_r + \Omega r\,\mathbf{e}_\varphi$ is $N\Omega r = 2m\Omega^2 r\dot{r}$ (the vertical part $mg\,\mathbf{k}$ is orthogonal to $\mathbf{v}$), exactly $\frac{d}{dt}\left[\frac{1}{2}m(\dot{r}^2 + \Omega^2 r^2)\right]$ with $\ddot{r} = \Omega^2 r$.

KEY POINTS

- FALSE.
- Coriolis force $-2m\boldsymbol{\Omega} \times \mathbf{v}_{rel} = -2m\Omega\dot{r}\,\mathbf{e}_\varphi \perp \mathbf{v}_{rel}$: zero power, always.
- The work comes from the centrifugal force $m\Omega^2 r\,\mathbf{e}_r$: $\frac{1}{2}m\dot{r}^2 = \frac{1}{2}m\Omega^2(r^2 - r_0^2)$, $r = r_0\cosh\Omega t$.
- The rod reaction $N\mathbf{e}_\varphi + mg\,\mathbf{k}$, $N = 2m\Omega\dot{r}$: its horizontal part balances the Coriolis force in S_1 and its vertical part the weight; in S_0 it does work at the rate $2m\Omega^2 r\dot{r}$.

Part (b) · 1 point

Free particle in an accelerating frame

The frame S_1 translates without rotating, with constant acceleration $\mathbf{a}_{O_1} = a_0\,\mathbf{e}$ ($a_0 > 0$, $\mathbf{e}$ a fixed unit vector) with respect to S_0. A particle subject to no real force (no gravity) is observed from S_1. Statement: "The only inertia force in S_1 is $-ma_0\,\mathbf{e}$, and the relative trajectory of the particle is a straight line only when its initial relative velocity is zero or parallel to $\mathbf{e}$; otherwise it is a parabola."

Step 1. Inertia forces

For a frame in pure translation, $\mathbf{a}_{abs} = \mathbf{a}_{O_1} + \mathbf{a}_{rel}$ (Notes §3.2 with $\boldsymbol{\omega} = \mathbf{0}$), so $m\mathbf{a}_{rel} = \mathbf{F} - m\mathbf{a}_{O_1}$: the only inertia force is $-ma_0\,\mathbf{e}$, uniform and constant. With $\mathbf{F} = \mathbf{0}$: $\mathbf{a}_{rel} = -a_0\,\mathbf{e}$.

Step 2. Relative motion

Integrating twice with Barrow's rule from 0 to t: $\mathbf{v}_{rel} = \mathbf{v}_0 - a_0 t\,\mathbf{e}$ and $\mathbf{r}_{rel} = \mathbf{r}_0 + \mathbf{v}_0 t - \frac{1}{2}a_0 t^2\,\mathbf{e}$. The path is straight exactly when $\mathbf{v}_{rel}$ keeps its direction, i.e. when $\mathbf{v}_{rel} \times \mathbf{a}_{rel} = \mathbf{0}$ (zero curvature, Notes §2.5):

$$\mathbf{v}_{rel} \times \mathbf{a}_{rel} = (\mathbf{v}_0 - a_0 t\,\mathbf{e}) \times (-a_0\,\mathbf{e}) = -a_0\,\mathbf{v}_0 \times \mathbf{e}$$

which vanishes if and only if $\mathbf{v}_0 = \mathbf{0}$ or $\mathbf{v}_0 \parallel \mathbf{e}$.

Step 3. The parabola

With $\mathbf{e} = \mathbf{i}$ and $\mathbf{v}_0 = U\mathbf{j}$: $x = x_0 - \frac{1}{2}a_0 t^2$, $y = y_0 + Ut$. Eliminating $t = (y - y_0)/U$ gives $x - x_0 = -\dfrac{a_0}{2U^2}(y - y_0)^2$, a parabola. Both claims hold: TRUE. (In S_0 the same particle moves in a straight line at constant velocity.)

KEY POINTS

- TRUE.
- $\mathbf{F}_{in} = -m\,\mathbf{a}_{O_1} = -ma_0\,\mathbf{e}$; $\mathbf{r}_{rel} = \mathbf{r}_0 + \mathbf{v}_0 t - \frac{1}{2}a_0 t^2\,\mathbf{e}$.
- $\mathbf{v}_{rel} \times \mathbf{a}_{rel} = -a_0\,\mathbf{v}_0 \times \mathbf{e}$ (constant): the path is straight if and only if $\mathbf{v}_0 \times \mathbf{e} = \mathbf{0}$.
- Example $\mathbf{v}_0 = U\mathbf{j}$, $\mathbf{e} = \mathbf{i}$: $x - x_0 = -\dfrac{a_0}{2U^2}(y - y_0)^2$.

Interpretation and checks

- **Dimensional check:** $m\Omega\dot{r}$ and ma_0 are N; $m\Omega^2 r^2$ is J; $a_0 y^2/U^2$ is m.
- **Limits:** $\Omega \to 0$ in (a) gives $r \equiv r_0$ (no driving force). $a_0 \to 0$ in (b) makes S_1 inertial and every free trajectory straight.
- **Common mistakes in the exam:** (1) attributing work to the Coriolis force (it is always normal to $\mathbf{v}_{rel}$); (2) adding centrifugal or Coriolis forces in a frame that only translates; (3) "a free particle moves in a straight line" in a non-inertial frame.